Are Headtracking Measures an Indicator of Depressive Symptoms?

Report 2: Advanced Statistical Analyses

Parv Shah (2025201093) Hardik (2025201046) Gaurav (2025201065)
Gujju Gang — Behavioural Research & Statistical Methods (BRSM)

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1 Introduction and Background

Depression is a global public health challenge affecting approximately 332 million people worldwide World Health Organization, 2021. A hallmark of Major Depressive Disorder (MDD) is *psychomotor retardation* — reduced speed and amplitude of voluntary movement that is directly observable in gait, speech and head movement (Bennabi et al., 2013).

360-degree (360°) video viewed in virtual reality (VR) offers a naturalistic context in which psychomotor retardation can be measured unobtrusively. Because the observer must physically rotate their head to explore the scene, passive head-tracking data capture the *quantity* and *speed* of voluntary environmental scanning. A depressed individual, driven by blunted affect and diminished motivation, would be predicted to rotate their head less and more slowly than a non-depressed counterpart.

Srivastava et al. (2025) reported compelling evidence for this hypothesis: in a 360° VR paradigm, individuals classified as moderately-to-severely depressed (PHQ-9 $\geq$ 10) showed substantially lower mean head-rotation speed than controls, after covarying for anxiety (ANCOVA: $p < 0.001$, $\eta^2 = .295$). This result points toward the potential of low-cost VR head-tracking as a complementary screening tool for depression in settings where clinical interview is impractical.

The present study attempts to replicate and extend this finding in a sample of $N = 40$ university students who participated in a 360° VR experiment as part of a behavioural research course. Participants watched five emotionally varied 360° videos while wearing a VR headset; head-tracking data were recorded continuously. Questionnaire measures of depression (PHQ-9), generalised anxiety (GAD-7), and trait anxiety (STAI-T) were collected, along with momentary affect (PANAS pre/post) and per-video subjective ratings of valence, arousal, and presence.

Report 1 established the descriptive and preliminary inferential landscape: the sample was predominantly non-clinical (80% below the PHQ-9 clinical cutoff), depression and anxiety were strongly correlated ($r = .58$, $p < .001$), mean rotation speed varied markedly across videos, and no group comparison survived Bonferroni correction. Report 1 also identified key methodological challenges the severely unbalanced depression groups ($n_{\text{dep}} = 8$, $n_{\text{non-dep}} = 32$) and the high depression-anxiety covariance as likely reasons for non-replication, and proposed a suite of advanced analyses for Report 2.

Research questions for Report 2:

RQ1. Does head-rotation speed differ significantly across the five 360° videos, and which pairs of videos drive that effect?

RQ2. Does PHQ-9 score (treated as a continuous predictor) predict mean head-rotation speed, both before and after covarying for anxiety (GAD-7, STAI-T)?

RQ3. Does depression moderate the subjective experience of VR (valence, arousal, presence ratings) or PANAS affect change?

RQ4. Does VR presence moderate the relationship between depression and head-movement?

2 Methods

2.1 Participants and Design

A convenience sample of $N = 40$ postgraduate students (36 male, 4 female; $M_{\text{age}} = 22.8$, $SD = 1.8$, range 19–27) completed the experiment. Participants were classified by PHQ-9 into a *depressed* group (PHQ-9 ≥ 10; $n = 8$, 20%) and a *non-depressed* group (PHQ-9 < 10; $n = 32$, 80%). Sixteen participants reported prior VR experience (40%).

The design was a within-subjects repeated measures study: every participant watched all five 360° videos ($k = 5$) in a fixed order, generating $N \times k = 200$ head-tracking observations. Depression group (or continuous PHQ-9 score) was the between-subjects factor of primary interest.

2.2 Materials and Measures

Five 360° videos were presented via a VR headset. Three were short (V1–V3: ≈ 60 s each) and two were longer (V4–V5: ≈ 3 min each). They varied systematically in emotional valence and arousal (Table 1).

Table 1: Video stimuli: descriptive ratings (Mean ± SD; $N = 40$).

ID	Video	Valence	Arousal	Presence
V1	Abandoned Buildings	5.20 (1.67)	4.60 (2.22)	29.60 (4.41)
V2	Beach	6.53 (1.75)	4.65 (2.03)	27.25 (5.31)
V3	Campus	6.15 (1.61)	4.33 (2.16)	26.95 (4.52)
V4	Horror (Nun)	3.67 (1.85)	5.95 (1.68)	19.85 (5.44)
V5	Tahiti Surf	7.12 (1.68)	6.50 (1.69)	28.02 (4.79)

Clinical measures: **PHQ-9** (Patient Health Questionnaire; Kroenke et al. 2001), **GAD-7** (Generalised Anxiety Disorder scale; Spitzer et al. 2006), and **STAI-T** (State-Trait Anxiety Inventory—Trait form; Spielberger et al. 1983). Affect was measured with the **PANAS** (Positive and Negative Affect Schedule; Watson et al. 1988) administered before and after the VR session. VR sickness was assessed with the **VRISE** scale (Kourtesis et al., 2019).

2.3 Head-Tracking Feature Extraction

Each raw head-tracking file contained time-stamped rotation data. From each file the following summary features were extracted per participant per video:

- **Mean rotation speed (total)**: mean of `RotationSpeedTotal` (°/s), the magnitude of instantaneous angular velocity, serving as the primary dependent variable.
- Mean yaw (horizontal) speed and mean pitch (vertical) speed.
- Standard deviation of total rotation speed (variability in scanning).
- Total angular range explored: sum of the peak-to-trough range on the X, Y, and Z rotation axes.
- **SDY (Standard Deviation of Yaw position)**: the SD of the cumulative yaw (horizontal) angular position over the video, measuring how much participants deviated from their mean looking direction. This was the key significant measure in Srivastava et al. (2025).

2.4 Statistical Analyses

The α level was set at .05 two-tailed throughout; multiple comparison corrections are stated where applied. Effect sizes are reported as Cohen's d, r (rank-biserial correlation), η_g^2 (generalised eta-squared for repeated measures), or standardised β as appropriate.

Analysis 1 Video main effect (RQ1). Mauchly's test of sphericity was conducted prior to the repeated-measures ANOVA. The result ($W = 0.74$, $p = .257$) was non-significant, confirming that the sphericity assumption was met; the standard uncorrected F and degrees of freedom are therefore reported. A one-way within-subjects ANOVA was then performed on mean total rotation speed with video (V1–V5) as the within-subjects factor. Significant main effects were followed by pairwise t-tests with Holm correction. Two non-parametric complements were used: (a) a Kruskal-Wallis test with pairwise Mann-Whitney U post-hoc tests (Bonferroni corrected) to confirm the between-video differences without distributional assumptions; and (b) Friedman's ANOVA, the non-parametric repeated-measures alternative, to corroborate the RM-ANOVA result under a within-subjects rank framework.

Analysis 2 Simple linear regression (RQ2). For each video separately, mean rotation speed was regressed on the continuous PHQ-9 score. The five resulting p-values were adjusted using both Holm and Benjamini-Hochberg False Discovery Rate (BH-FDR) procedures.

Analysis 3 ANCOVA with anxiety covariates (RQ2). For each video, an OLS regression model was fitted: $\text{speed}_i = \beta_0 + \beta_1 \text{PHQ}_{z,i} + \beta_2 \text{GAD}_{z,i} + \beta_3 \text{STAI}_{z,i} + \varepsilon_i$, where subscript z denotes z-standardised predictors. Type II ANOVA partitioning was used to isolate the unique variance attributable to each predictor. Unstandardised β coefficients with 95% confidence intervals are reported.

Analysis 4 Linear Mixed Model (RQ3). To exploit the full repeated-measures structure while treating PHQ-9 continuously, a series of LMMs were fitted using Restricted Maximum Likelihood (REML) for parameter estimation and ML for model comparison via AIC/BIC:

- **Model 1**: Video fixed effect only, random participant intercept.
- **Model 2**: PHQ-9(z) + video, random participant intercept.

- **Model 3**: PHQ-9(z) + GAD-7(z) + video, random participant intercept.
- **Model 4**: PHQ-9(z) $\times$ video interaction + random participant intercept.

The intraclass correlation (ICC) was computed as $\sigma^2_{\text{between}}/(\sigma^2_{\text{between}} + \sigma^2_{\text{within}})$ to quantify individual-level clustering.

Analysis 5 Post-hoc power analysis (RQ4). Observed power was computed for the original effect size reported by Srivastava et al. (2025) ($\eta^2 = .295$, equivalent $d = 1.29$) given the current design ($n_{\text{dep}} = 8$, $n_{\text{non-dep}} = 32$, $\alpha = .05$, two-tailed). The minimum detectable effect (MDE) at 80% power was also computed.

Analysis 6 Subjective experience and PANAS (RQ5). LMMs were used to test whether PHQ-9 predicted video-level valence, arousal, and presence ratings. Wilcoxon signed-rank tests examined pre-to-post PANAS changes within each depression group; Mann-Whitney U tests compared change scores between groups.

Analysis 7 Multiple Linear Regression with interaction term (RQ6). For each video, a multiple linear regression model was fitted with PHQ-9(z), presence(z), and their product term as predictors of mean rotation speed. The product term (PHQz $\times$ Presence$z$) tests whether presence moderates the depression–headtracking relationship (i.e., whether the PHQ-9 slope differs by level of immersion).

3 Results

3.1 Video Main Effect on Head-Rotation Speed (RQ1)

A one-way repeated-measures ANOVA revealed a large, highly significant effect of video on mean head-rotation speed ($F(4, 156) = 34.58$, $p < .001$, $\eta^2_g = .181$; Figure 1). Mauchly's test confirmed sphericity was met ($W = 0.74$, $p = .257$), so the standard uncorrected F is reported. This indicates that the type of 360° content viewed accounts for approximately 18% of the total variance in head-movement speed.

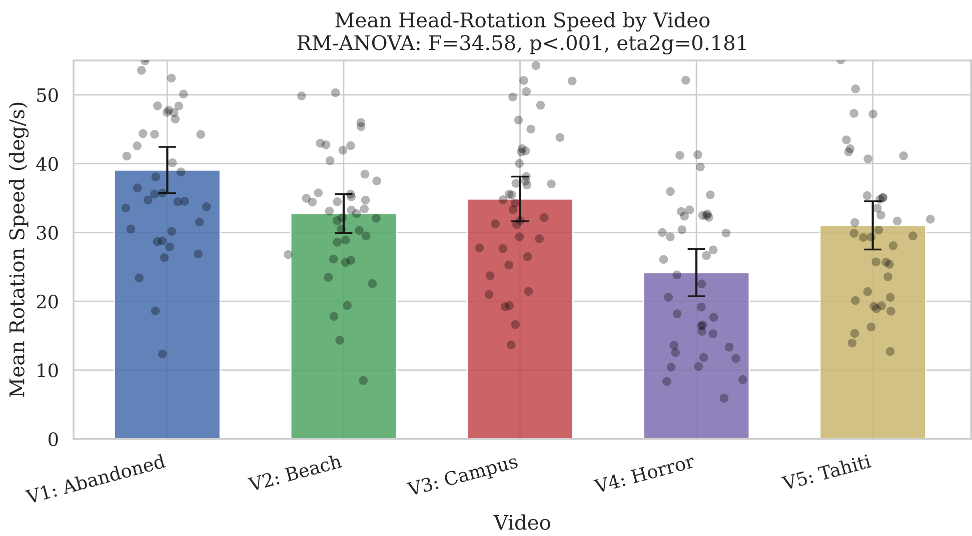

Figure 1: Mean head-rotation speed (°/s) by video. Bars show group means with 95% CIs; dots are individual participant values. RM-ANOVA (sphericity met, Mauchly $W = 0.74$, $p = .257$): $F(4, 156) = 34.58$, $p < .001$, $\eta^2_g = .181$.

Post-hoc comparisons (paired t-tests, Holm correction; Table 2) revealed a clear pattern: V4 (Horror) had markedly lower mean rotation speed than all other videos (all $p < .001$, $d > 0.84$). V1 (Abandoned Buildings) had significantly higher speed than V2 (Beach; $p < .001$, $d = 0.63$), V3 (Campus; $p = .002$, $d = 0.39$), V4 ($p < .001$, $d = 1.35$), and V5 (Tahiti; $p < .001$, $d = 0.72$). V3 also differed significantly from V5 ($p = .014$, $d = 0.35$). V2 and V3 did not differ from each other ($p = .195$).

These results were confirmed by two non-parametric tests. The Kruskal-Wallis test yielded $H(4) = 33.51$, $p < .001$, $\eta^2 = .151$; pairwise Mann-Whitney U post-hoc tests (Bonferroni corrected) found 4 of 10 pairs significant, with V4 driving most differences (Figure 2). Friedman's ANOVA — the non-parametric repeated-measures alternative — confirmed the video effect: $\chi^2(4, N = 40) = 70.96$, $p < .001$, Kendall's $W = 0.444$ (large effect).

Table 2: Pairwise comparisons (RM-ANOVA, Holm-corrected paired t-tests, $df = 39$).

Pair	t	p_{unc}	p_{Holm}	Hedges' g
V1 vs V2	5.48	$< .001$	$< .001$	0.63
V1 vs V3	3.85	$< .001$	.002	0.39
V1 vs V4	11.23	$< .001$	$< .001$	1.35
V1 vs V5	6.20	$< .001$	$< .001$	0.72
V2 vs V3	-1.70	.097	.195	-0.21
V2 vs V4	5.81	$< .001$	$< .001$	0.84
V3 vs V4	6.65	$< .001$	$< .001$	0.98
V3 vs V5	2.99	.005	.014	0.35
V4 vs V5	-5.26	$< .001$	$< .001$	-0.61

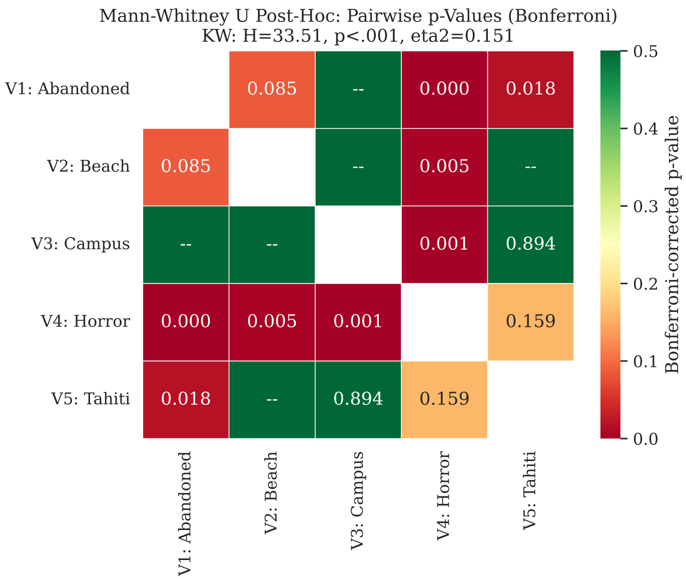

Figure 2: Pairwise Mann-Whitney U post-hoc p-values (Bonferroni-corrected) from the Kruskal-Wallis test. Green cells indicate significant p-values; red indicates non-significant comparisons. Four pairs were significant: V1–V4, V1–V5, V2–V4, and V3–V4.

The depressed-Horror (V4) pattern is consistent with a *freezing or hypervigilance response*: when confronted with an aversive, threatening stimulus, participants minimise exploratory scanning to attend to the perceived threat. This effect appeared across all participants and was not modulated by depression group status.

3.2 PHQ-9 as a Continuous Predictor of Head-Rotation Speed (RQ2)

Simple linear regression. Regressing mean rotation speed on PHQ-9 scores separately for each video produced uniformly non-significant and negligible effects (Table 3; Figure 3). All β coefficients were small in magnitude ($|\beta| < 0.35$ °/s per unit PHQ-9) and all $R^2 < .02$. After applying both Holm and BH-FDR corrections, no test remained significant.

Table 3: Simple regression of PHQ-9 on mean rotation speed per video ($N = 40$).

Video	β	95% CI	p	p_{FDR}	R^2
V1: Abandoned	−0.18	[−0.95, 0.59]	.636	.795	.006
V2: Beach	0.19	[−0.45, 0.84]	.544	.795	.010
V3: Campus	−0.23	[−0.97, 0.51]	.534	.795	.010
V4: Horror	−0.03	[−0.82, 0.75]	.933	.933	.000
V5: Tahiti	−0.35	[−1.14, 0.45]	.379	.795	.020

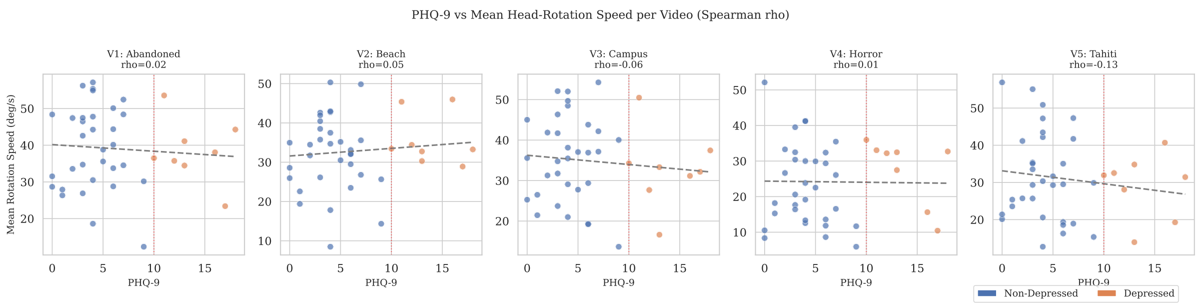

Figure 3: Scatter plots of PHQ-9 score against mean head-rotation speed (°/s) per video. Blue dots = non-depressed; orange dots = depressed. Dashed line = OLS regression. Red dotted vertical line = clinical cutoff (PHQ-9 = 10). Spearman ρ is annotated above each panel (* = $p < .05$ uncorrected). No panel reaches significance after correction.

ANCOVA controlling for anxiety. Adding GAD-7 and STAI-T as covariates did not reveal a suppressed PHQ-9 effect; on the contrary, all models showed negative adjusted R^2, indicating that the predictors collectively do not explain head-tracking variance beyond chance (Table 4; Figure 4). Type II ANOVA partitioning confirmed that neither PHQ-9 nor the anxiety covariates contributed uniquely significant variance in any video.

Table 4: ANCOVA: PHQ-9 effect on mean rotation speed, controlling for GAD-7 and STAI-T ($N = 40$, standardised predictors).

Video	β_{PHQ}	95% CI	p	p_{FDR}	Adj. R^2
V1: Abandoned	−2.07	[−6.86, 2.72]	.387	.646	−.051
V2: Beach	1.28	[−2.72, 5.28]	.521	.651	−.042
V3: Campus	−2.20	[−6.78, 2.39]	.339	.646	−.033
V4: Horror	0.10	[−4.79, 4.99]	.967	.967	−.048
V5: Tahiti	−2.61	[−7.60, 2.37]	.295	.646	−.046

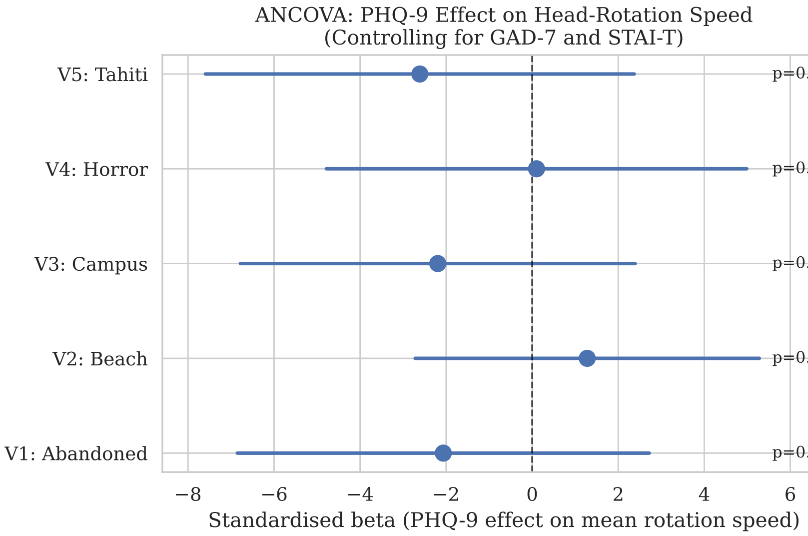

Figure 4: Forest plot of standardised PHQ-9 β coefficients from ANCOVA (controlling for GAD-7 and STAI-T) with 95% confidence intervals. All intervals span zero; no video shows a significant PHQ-9 effect.

3.3 Linear Mixed Model (RQ3)

The LMM structure allowed the full within-subjects repeated-measures design to be modelled simultaneously, with participant as a random intercept, removing individual speed baselines from the error term.

The ICC was 0.694, indicating that approximately **69% of the total variance in rotation speed is attributable to stable individual differences** between participants (Table 5). This is a high ICC and confirms that individuals differ substantially in their natural head-movement style a baseline variability that obscures any depression-related signal.

Table 5: Linear Mixed Model results: fixed effects for the PHQ-9 predictor.

Model	β_{PHQ}	SE	p	AIC
Model 2: PHQ + video	−0.55	1.47	.710	1388.2
Model 3: PHQ + GAD + video	−1.20	1.81	.508	1389.8
Model 4: PHQ × video	—	—	—	1391.9

Model comparison via AIC indicates that the simplest model (Model 2) is preferred over models adding GAD-7 or the PHQ-9 $\times$ video interaction term ($\Delta\text{AIC} \leq 3.7$). In all models, PHQ-9 showed a non-significant, near-zero association with head-rotation speed: $\beta = -0.55$ (SE = 1.47), $p = .71$ (Model 2); $\beta = -1.20$ (SE = 1.81), $p = .51$ (Model 3). The video fixed effects were significant across all models (all $p < .001$), confirming the robust content-driven speed variation observed in Section 3.1.

Figure 5 displays the fixed-effect estimates and the distribution of individual random intercepts. The spread of random intercepts ($\sigma_{\text{between}} = \sqrt{79.12} \approx 8.9°/\text{s}$) is large relative to the residual ($\sigma_{\text{within}} = \sqrt{34.85} \approx 5.9°/\text{s}$), further illustrating the dominance of individual differences.

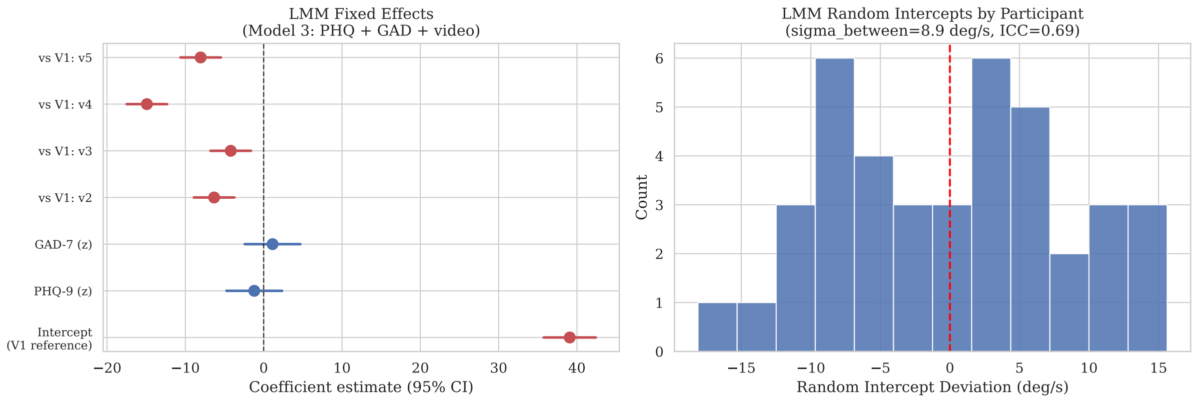

Figure 5: Left: Fixed-effect estimates with 95% CIs from LMM Model 3 (PHQ + GAD + video). Dashed line at 0. Right: Histogram of participant-level random intercepts. The wide spread (SD $\approx 8.9°/\text{s}$) confirms large individual differences in baseline scanning speed (ICC = 0.694).

3.4 Multiple Comparison Corrections (MCC)

Report 1 applied a conservative Bonferroni correction to 25 Mann-Whitney tests (5 videos $\times$ 5 head-tracking measures), which resulted in no significant findings. In Report 2, a sixth measure **SDY (standard deviation of yaw angular position)**was added because it was the key significant finding in Srivastava et al. (2025). This expanded the family to 30 tests (5 videos $\times$ 6 measures). Three correction methods were compared (Figure 6):

Table 6: Significant results ($p < .05$) under different MCC methods ($k = 30$ tests).

Correction Method	Significant Tests
None (uncorrected)	3 / 30
Bonferroni	0 / 30
Holm (step-down Bonferroni)	0 / 30
Benjamini-Hochberg FDR	0 / 30

Even the most liberal BH-FDR procedure, which controls the expected proportion of false discoveries rather than the family-wise error rate, yielded zero significant results. The three uncorrected nominal findings including SDY for the horror video (V4: $p = .025$

uncorrected) are therefore attributable to Type I error at the expected rate ($\alpha = .05$, expected false positives $= 30 \times .05 = 1.5$). Notably, SDY was the primary significant measure in Srivastava et al. (2025), yet our data show no reliable group difference even before correction (4 of 5 videos $p > .70$). This convergence across all correction methods and all six measures strengthens the conclusion that there are no genuine differences in head-tracking behaviour between depression groups in the present sample.

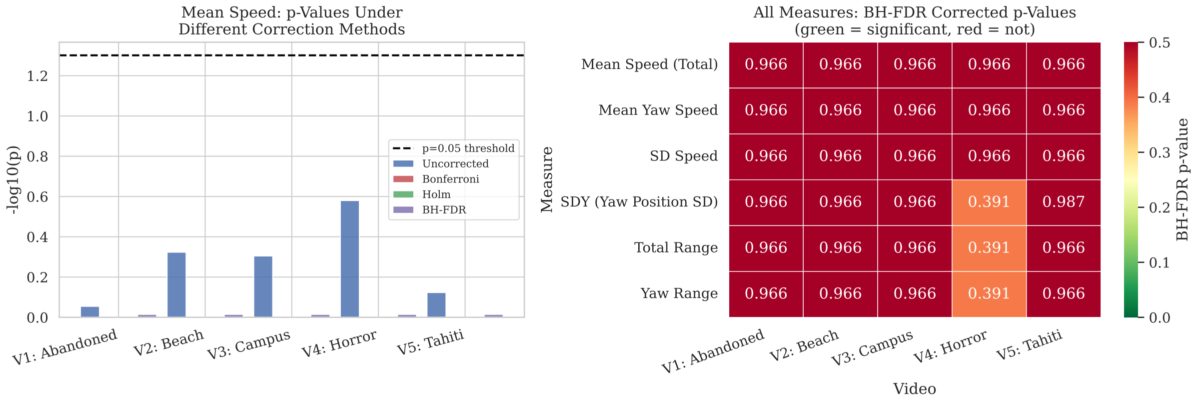

Figure 6: Left: $-\log_{10}(p)$ for mean speed per video under four MCC methods. Dashed line $= -\log_{10}(0.05)$. No bar clears the threshold after any correction. Right: Heatmap of BH-FDR corrected p-values for all six head-tracking measures (including SDY) across videos.

3.5 Post-Hoc Power Analysis (RQ4)

The power of the current design to detect the effect originally reported by Srivastava et al. (2025) was computed. Converting $\eta^2 = .295$ to Cohen's d yields $d = 1.29$. Despite the unbalanced design ($n_{\text{dep}} = 8$, $n_{\text{non-dep}} = 32$), the current study had **89.1% power** to detect this large effect (Figure 7).

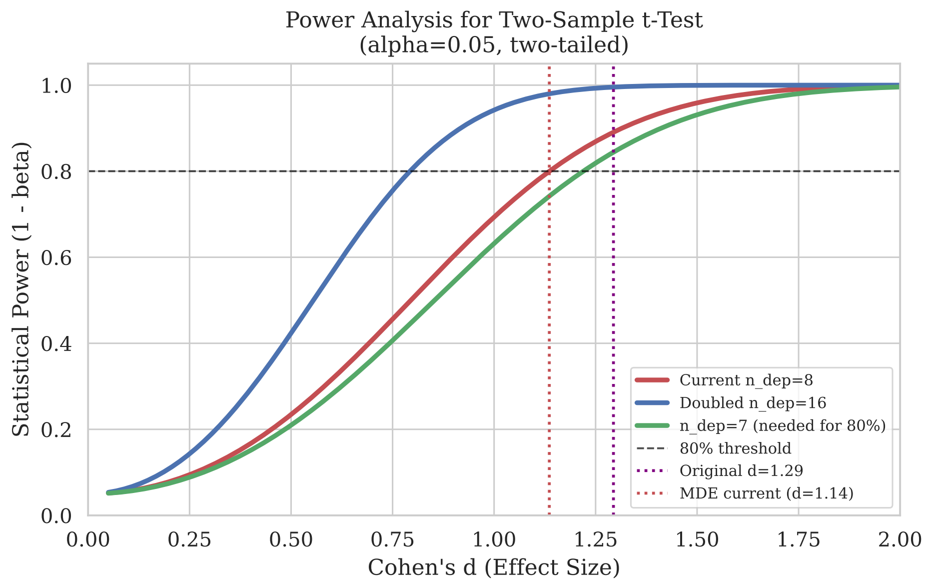

Figure 7: Power curves for the current design ($n_{\text{dep}} = 8$) and two larger designs. The current design (red curve) achieves 89% power at the original effect size ($d = 1.29$, purple dotted line) but requires $d = 1.14$ to achieve 80% power (red dotted MDE line). This rules out the possibility that non-replication is due to low power for effects of the reported magnitude.

Interpretation: The high power (89%) to detect the original large effect rules out insufficient power as the primary explanation for non-replication. The minimum detectable effect is $d = 1.14$ the current design can only detect very large effects. For smaller, more realistic effect sizes ($d = 0.50$), power drops to only 23%. This means that if depression has a small-to-moderate effect on head-tracking in student populations, the present study is substantially underpowered to detect it.

3.6 PHQ-9 and Subjective VR Experience (RQ5)

Video ratings. Across all videos combined, LMMs revealed that PHQ-9 was not associated with valence ($\beta = -0.03$, 95% CI $[-0.32, 0.26]$, $p = .832$) or presence ratings ($\beta = 0.11$, 95% CI $[-0.99, 1.21]$, $p = .841$). However, PHQ-9 **significantly predicted lower arousal ratings** ($\beta = -0.44$, 95% CI $[-0.78, -0.10]$, $p = .012$): each unit increase in PHQ-9 was associated with 0.44-point lower arousal on a 9-point scale. This suggests that while depression does not alter perceived pleasantness or sense of immersion, it blunts the perceived arousal or emotional intensity of video content consistent with anhedonia and emotional blunting in MDD (Figure 8).

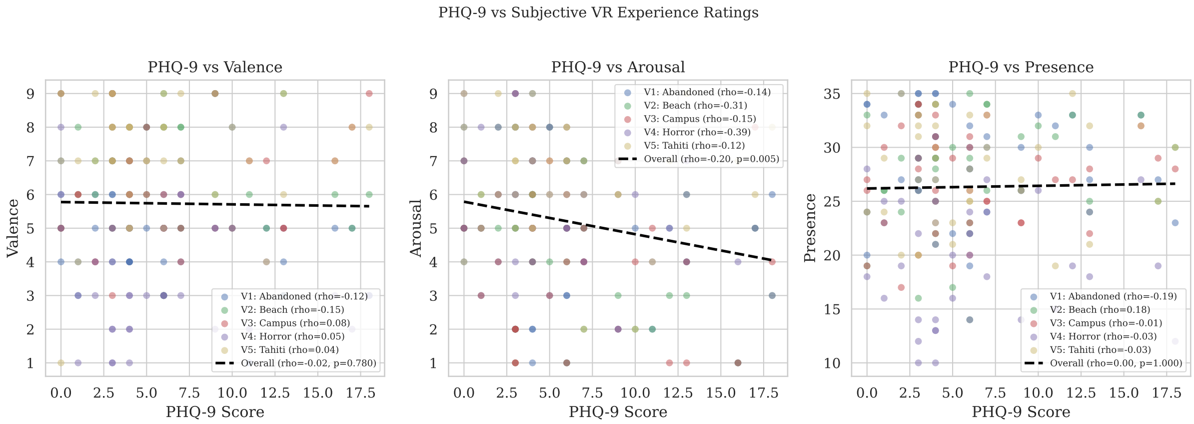

Figure 8: PHQ-9 vs. subjective ratings of valence, arousal, and presence across all videos. Each colour represents a different video. The overall regression line (black dashed) is fitted across all observations. Only arousal shows a significant negative association ($p = .012$).

PANAS affect change. Splitting the PANAS analysis by depression group revealed a divergent pattern (Figure 9):

- **Depressed group** ($n = 8$): A statistically significant *decrease* in positive affect was observed post-VR ($M_{\text{pre}} = 29.62$, $M_{\text{post}} = 25.38$, $\Delta = -4.25$; Wilcoxon $W = 0$, $p = .008$, $d = -1.54$), a large effect. Negative affect did not change significantly ($p = .547$).
- **Non-Depressed group** ($n = 32$): No significant change in positive affect ($M_{\text{pre}} = 35.75$, $M_{\text{post}} = 33.84$, $\Delta = -1.91$; $p = .247$) or negative affect ($p = .607$).

The between-group comparison of PA change did not reach significance after accounting for the modest sample sizes ($U = 80$, $p = .107$). These results suggest that for individuals with elevated depressive symptoms, VR exposure produced a clinically meaningful reduction in positive affect.

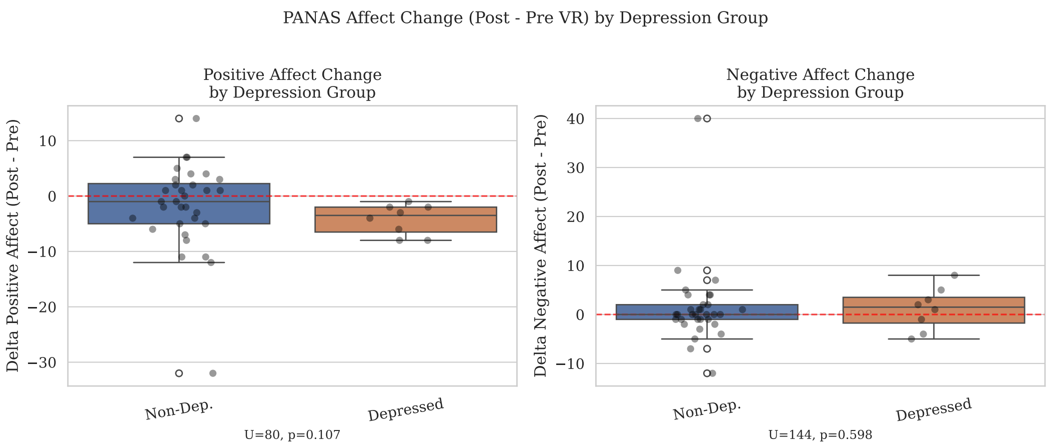

Figure 9: PANAS affect change (Post − Pre VR) by depression group. Boxplots with individual participant overlaid as dots. The depressed group showed a significant decrease in positive affect ($p = .008$, $d = -1.54$); the non-depressed group did not.

3.7 Presence as a Moderator (RQ6)

Moderation analyses found no evidence that VR presence moderated the relationship between PHQ-9 and head-rotation speed in any of the five videos (all interaction $p > .27$; Table 7). The adjusted R^2 values were negligible for most videos. Thus, VR immersion does not appear to buffer or amplify any latent depression-related scanning differences.

Table 7: Moderation analysis: PHQ-9 × Presence interaction on mean rotation speed.

Video	$\beta_{\text{interaction}}$	p	Adj. R^2
V1: Abandoned	−1.02	.558	.016
V2: Beach	1.35	.490	−.033
V3: Campus	1.68	.448	−.017
V4: Horror	−1.94	.271	−.030
V5: Tahiti	−0.64	.700	.199

4 Conclusion

This study set out to test whether head-tracking behaviour during 360° VR exposure is a reliable marker of depressive symptoms. The main findings can be summarised as follows:

1. **Video content strongly drives head-rotation speed.** The repeated-measures ANOVA revealed a large, robust video effect ($F(4, 156) = 34.58$, $p < .001$, $\eta_g^2 = .18$; sphericity met, $W = 0.74$, $p = .257$), with the horror video (V4) producing markedly less movement and the exploratory abandoned-buildings video (V1) producing the most. Both the Kruskal-Wallis test ($H(4) = 33.51$, $\eta^2 = .151$) and Friedman's ANOVA ($\chi^2(4) = 70.96$, Kendall's $W = .44$) converged on the same conclusion.

2. **Depression (PHQ-9) does not predict head-tracking behaviour.** Across all analytical approaches simple regression, ANCOVA with anxiety covariates, and LMM PHQ-9 showed near-zero and uniformly non-significant associations with head-rotation speed. Critically, SDY (standard deviation of yaw position), the key significant measure in Srivastava et al. (2025), also showed no reliable group difference in the present sample ($p > .70$ in 4 of 5 videos; V4: $p = .025$ uncorrected, $p = .74$ after Bonferroni). After both Holm and BH-FDR corrections across all 30 tests (6 measures × 5 videos), no test survived.

3. **Large individual differences dominate the data.** The ICC of 0.694 indicates that people differ substantially in their natural scanning style, and these differences are large relative to any depression-related signal. Future studies should either collect a pre-VR baseline head-tracking measure to residualise individual differences, or use change scores.

4. **Depression predicts blunted arousal perception and reduced positive affect.** PHQ-9 significantly predicted lower arousal ratings of video content ($\beta = -0.44$, $p = .012$). Within the depressed group, VR exposure caused a large decrease in positive affect ($d = -1.54$), whereas this effect was absent in non-depressed participants. These findings are consistent with the literature on anhedonia and reduced reward sensitivity in depression.

Limitations. The most critical limitations are (a) the small and severely unbalanced depression groups ($n_{\text{dep}} = 8$), which restrict power for detecting small-to-moderate effects; (b) the predominantly male, young, postgraduate sample, which limits external validity; (c) the predominantly sub-clinical range of depression scores (80% below PHQ-9 ≥ 10); and (d) the inability to partial out habitual individual differences in head-movement due to lack of a baseline condition.

Author Contributions

The complete code and analysis pipeline for this project are at: GitHub repository

Parv Shah (2025201093) led the conceptualisation and design of the analytical framework for Report 2, conducted all primary statistical analyses (repeated-measures ANOVA, ANCOVA, LMM, power analysis and moderation analyses), produced all figures and drafted the report.

Gaurav (2025201065) contributed to data preprocessing and head-tracking feature extraction, assisted in the interpretation of results, performed verification of key statistical outputs and provided substantive feedback on successive drafts of the manuscript.

Hardik (2025201046) conducted the literature review and contextualised findings within the broader depression and VR literature, contributed to the discussion of methodological limitations and contributed to proofreading and final revision of the report.

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